

COMPANION SLIDES

Chapter 13

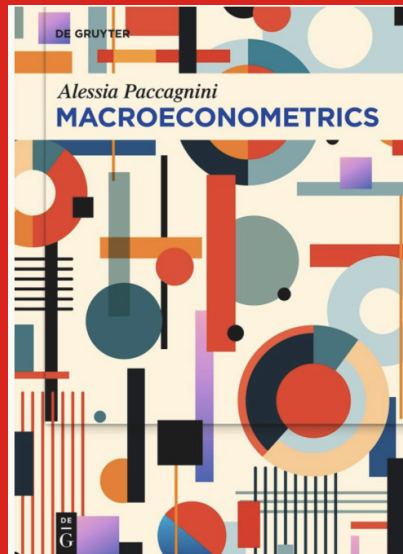
Multi-Country Methods

Policy analysis in an interconnected world

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Policy analysis across interconnected economies

How should we model monetary, fiscal and global shocks when countries, regions or states are **heterogeneous, dynamic and mutually connected**?

- ▶ Multi-unit data add observations and identifying variation.
- ▶ But units trade, share financial linkages and react to common global shocks.
- ▶ Naive pooling can therefore produce misleading inference.

The challenge is to exploit cross-sectional information without ignoring dependence.

Panel foundations

- ▶ Fixed and random effects
- ▶ Dynamic panels and Nickell bias
- ▶ Arellano–Bond and system GMM
- ▶ Panel VARs and heterogeneity

Global transmission and identification

- ▶ GVAR / VARX* models
- ▶ Weak exogeneity and stacking
- ▶ GIRFs and spillover networks
- ▶ Shift-share and granular IVs

Empirical illustrations: **PVAR oil-price shocks** and **GVAR US monetary-policy spillovers**.

- ▶ Explain Nickell bias and why dynamic fixed-effects estimators are biased when T is short.
- ▶ Implement difference and system GMM and interpret specification tests.
- ▶ Distinguish pooled, heterogeneous and interacted Panel VARs.
- ▶ Construct trade-weighted foreign variables and test weak exogeneity in a GVAR.
- ▶ Stack country-specific VARX* models into a global system.
- ▶ Compute GIRFs that trace shocks through international networks.
- ▶ Understand shift-share and granular IV identification strategies.
- ▶ Choose among Panel VAR, GVAR and identification-based approaches for a given policy question.

What we gain

- ▶ more observations
- ▶ policy timing variation
- ▶ heterogeneous exposure
- ▶ spillover measurement
- ▶ cross-country comparisons

What becomes harder

- ▶ unit heterogeneity
- ▶ serial correlation
- ▶ common shocks
- ▶ cross-sectional dependence
- ▶ identification across networks

Dependence is not always a nuisance: often it is the object of interest.

Method	Dependence	Identification	Typical use
Dynamic panel	time effects / robust inference	lagged values as IVs	persistent outcomes, average effects
Panel VAR	common or heterogeneous dynamics	recursive / sign restrictions	average dynamic responses
GVAR	explicit trade/financial weights	GIRF / structural	international transmission
Shift-share	heterogeneous exposure	exogenous shares or shocks	local effects of aggregate shocks
Granular IV	large-unit shocks	size-based instruments	micro shocks with aggregate effects

$$y_{it} = \alpha_i + x'_{it}\beta + \varepsilon_{it}$$

- ▶ α_i captures time-invariant unit heterogeneity.
- ▶ **Fixed effects**: allows correlation between α_i and regressors.
- ▶ **Random effects**: treats α_i as random and independent of regressors.
- ▶ The choice concerns assumptions about unobserved heterogeneity, not just efficiency.

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

- ▶ Demeaning removes time-invariant unit effects.
- ▶ Identification comes from **within-unit variation over time**.
- ▶ Time-invariant regressors disappear under the within transformation.

Caution

Panel inference must still address serial correlation, heteroskedasticity and cross-sectional dependence.

FE

- ▶ robust to correlation with α_i
- ▶ uses within variation
- ▶ less restrictive

RE

- ▶ more efficient if assumptions hold
- ▶ uses within and between variation
- ▶ requires orthogonality

The **Hausman test** compares the estimators under the null that the RE orthogonality conditions are valid.

- ▶ Countries are exposed to common shocks and international linkages.
- ▶ Conventional standard errors can severely understate uncertainty.
- ▶ Cluster-robust methods help with within-unit dependence, but strong common factors require additional treatment.
- ▶ Cross-sectional dependence becomes central when the research question is about **spillovers**.

$$y_{it} = \gamma y_{i,t-1} + x'_{it}\beta + \alpha_i + \varepsilon_{it}$$

- ▶ Lagged outcomes capture persistence and adjustment dynamics.
- ▶ But $y_{i,t-1}$ is correlated with the transformed error after removing fixed effects.
- ▶ With small T , FE estimates of persistence suffer from **Nickell bias**.

Approximation

The bias in $\hat{\gamma}$ is of order $1/T$ and is often negative.

- ▶ First-difference the model to remove α_i .
- ▶ Use deeper lags of y as instruments for $\Delta y_{i,t-1}$.
- ▶ Anderson–Hsiao uses a limited instrument set.
- ▶ Arellano–Bond difference GMM exploits a larger collection of valid moment conditions.

$$E[y_{i,t-s} \Delta \varepsilon_{it}] = 0, \quad s \geq 2$$

Blundell–Bond idea

Combine equations in **differences** with equations in **levels**, adding moment conditions that can improve efficiency for persistent series.

- ▶ Particularly useful when lagged levels are weak instruments for differences.
- ▶ Requires additional assumptions about initial conditions.
- ▶ More instruments are not automatically better.

- ▶ **AR(1)/AR(2) tests**: check serial correlation in differenced residuals.
- ▶ **Hansen/Sargan tests**: assess overidentifying restrictions.
- ▶ Limit instrument count to avoid instrument proliferation.
- ▶ Report robustness to lag depth and instrument collapsing.

A very high Hansen p -value can be a warning sign, not a certificate of validity.

$$y_{it} = A_{0i} + A_{1i}y_{i,t-1} + \cdots + A_{pi}y_{i,t-p} + u_{it}$$

- ▶ Extends VAR logic to a panel of units.
- ▶ Captures dynamic feedback among multiple endogenous variables.
- ▶ Main modeling choice: how much dynamics should be pooled across units?

Pooled

Common coefficients across units.

Heterogeneous

Unit-specific coefficients; summarize across units.

Interacted

Dynamics depend on unit traits or foreign aggregates.

The right specification depends on whether **commonality, heterogeneity or spillovers** dominate the research question.

- ▶ GMM with Helmert / forward orthogonal deviations.
- ▶ Mean Group estimation: estimate unit models and average coefficients.
- ▶ Bayesian partial pooling: shrink unit-specific dynamics toward a common distribution.
- ▶ Identification can use recursive, sign or other structural restrictions.

- ▶ IRFs can summarize an average response or display the distribution across units.
- ▶ Heterogeneity is economically meaningful: transmission can vary with institutions, trade openness or financial structure.
- ▶ Structural identification assumptions must be credible across the units being compared.

Core idea

Model each country jointly with weighted averages of its trading or financial partners.

$$y_{it}^* = \sum_{j \neq i} w_{ij} y_{jt}$$

- ▶ w_{ij} can reflect bilateral trade, financial exposure or another network.
- ▶ Foreign aggregates transmit shocks across countries.
- ▶ The weighting matrix encodes the economic network.

$$y_{it} = \alpha_i + \Phi_i y_{i,t-1} + \Lambda_{i0} y_{it}^* + \Lambda_{i1} y_{i,t-1}^* + \varepsilon_{it}$$

- ▶ Domestic variables react to their own history and foreign conditions.
- ▶ Country models can differ in dimension and coefficients.
- ▶ A dominant unit may be modeled differently from smaller economies.

Key GVAR assumption

Foreign aggregates are treated as weakly exogenous for the long-run parameters of each country model.

- ▶ Small economies can react to global conditions without materially determining them contemporaneously.
- ▶ The assumption should be tested rather than simply asserted.
- ▶ Failure can signal that a unit is too influential or that the foreign-variable construction needs revision.



$$G_0 y_t = G_1 y_{t-1} + u_t \quad \Rightarrow \quad y_t = F_1 y_{t-1} + G_0^{-1} u_t$$

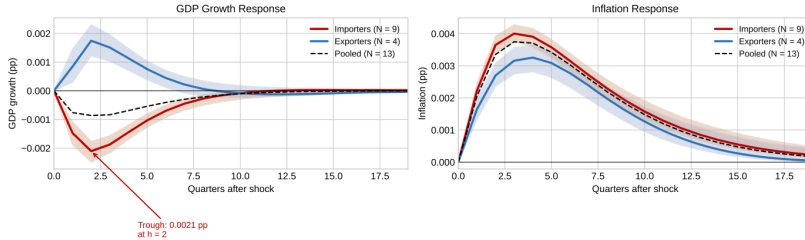
- ▶ GIRFs trace a shock while respecting the historical covariance structure of innovations.
- ▶ They do not require an arbitrary recursive ordering of all variables in a very large global system.
- ▶ Responses reveal both **direct effects** and **network propagation**.

A shock's global effect depends on both country dynamics and the network connecting countries.

	Panel VAR	GVAR
Main goal	average/heterogeneous dynamics	explicit cross-country transmission
Dependence	pooled, heterogeneous or interacted	network-weighted foreign variables
Strength	efficient comparison across units	maps international spillover channels
Typical question	How does policy affect countries differently?	How does a shock in one economy propagate globally?

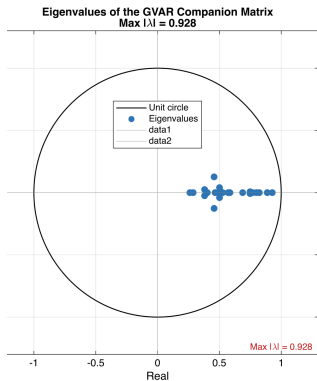
- ▶ 13 economies split into **4 oil exporters** and **9 importers**.
- ▶ Variables: GDP growth, CPI inflation and global oil-price growth.
- ▶ Sample: 1979Q2–2019Q4; interacted PVAR(2).
- ▶ Recursive ordering: $[\Delta y \rightarrow \pi \rightarrow \Delta p^{oil}]$.
- ▶ Companion exercise uses a **calibrated simulated dataset** for pedagogical purposes.

Panel VAR: Response to a Positive Oil Price Shock
Cholesky ordering: $[\Delta y \rightarrow \pi \rightarrow \Delta p^{oil}]$ | Shaded: 90% sieve-bootstrap CI



The
simulated PVAR exercise is designed to make core–periphery / exporter–importer heterogeneity visible
in dynamic responses.

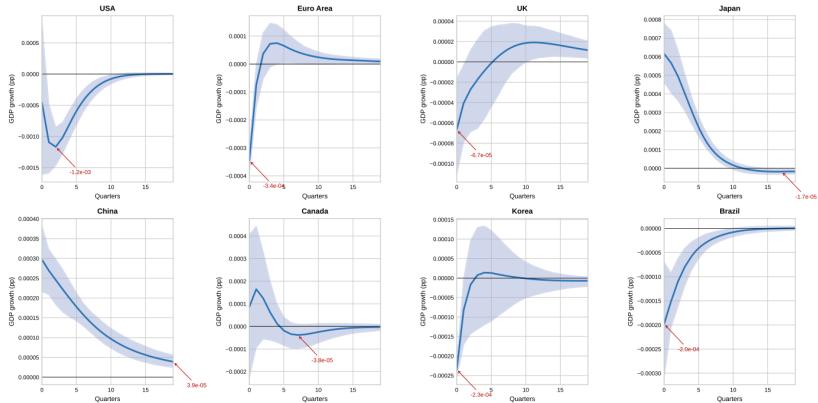
- ▶ 8 economies: USA, Euro Area, UK, Japan, China, Canada, Korea and Brazil.
- ▶ Domestic variables: GDP growth, CPI inflation and short-term interest rate.
- ▶ Non-US countries use VARX*(1,1); USA is treated as the dominant unit.
- ▶ Bilateral trade weights calibrate international exposure.
- ▶ Sample: 1979Q2–2019Q4; companion exercise uses calibrated simulated data.



Companion example

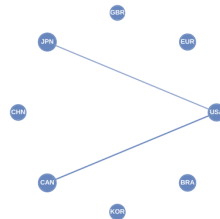
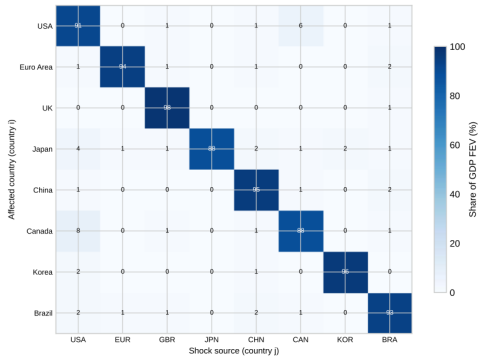
The maximum eigenvalue modulus is **0.928**, inside the unit circle.

- ▶ The system is dynamically stable.
- ▶ Near-unit-root behavior reflects persistent interest-rate dynamics.



GDP-growth responses to a one-standard-deviation contractionary US monetary-policy shock in the calibrated 8-country GVAR exercise.

- ▶ The US response is clearly contractionary, with the largest domestic decline appearing after the shock.
- ▶ Several economies show positive responses on impact because the GIRF conditions on the historical covariance structure of shocks.
- ▶ Contractionary spillovers emerge at longer horizons and differ substantially across countries.
- ▶ The corrected figure therefore highlights both **heterogeneous spillovers** and the distinction between generalized and orthogonal structural responses.



A network

view emphasizes that shock propagation depends on the pattern and strength of cross-country connections.

$$Z_i = \sum_k s_{ik} g_k$$

- ▶ s_{ik} : baseline exposure of unit i to sector/source k .
- ▶ g_k : aggregate shift affecting source k .
- ▶ Identification can rely on exogenous shares, exogenous shocks, or carefully stated combinations of assumptions.
- ▶ Inference must account for common exposure structures across units.

Granular idea

Idiosyncratic shocks to very large firms, banks or sectors need not wash out in the aggregate.

- ▶ Size concentration can create aggregate fluctuations from micro shocks.
- ▶ Properly isolated granular shocks can instrument endogenous aggregate relationships.
- ▶ Identification relies on assumptions distinct from shift-share designs.

- ▶ Multi-unit policy analysis often involves treatment at different dates.
- ▶ Effects can vary across units and over event time.
- ▶ Simple two-way fixed-effects summaries can be misleading under heterogeneous treatment effects.
- ▶ Research design and estimand should be explicit before estimation.

Ask first

- ▶ Average effect?
- ▶ Dynamic heterogeneity?
- ▶ Network spillovers?
- ▶ Causal exposure variation?

Then choose

- ▶ Dynamic panel
- ▶ Panel VAR
- ▶ GVAR
- ▶ Shift-share / granular IV

The research question should determine the architecture, not the other way around.

1. Define the unit, shock and transmission mechanism.
2. Diagnose heterogeneity and cross-sectional dependence.
3. Decide whether linkages are nuisance controls or substantive objects.
4. Select panel, PVAR, GVAR or exposure-based identification.
5. State and test the key identifying assumptions.
6. Compute dynamic responses and uncertainty measures.
7. Probe robustness to weights, samples, instruments and unit definitions.

1. Multi-unit data improve information but introduce dependence and heterogeneity.
2. Dynamic panels require methods that address lagged-outcome endogeneity.
3. Panel VARs are useful for common and heterogeneous dynamic responses.
4. GVARs make international linkages explicit and trace network propagation.
5. Credible identification depends on transparent assumptions about instruments, exposures and spillovers.

- ▶ Re-estimate a dynamic panel while varying the instrument set and report AR/Hansen diagnostics.
- ▶ Compare pooled, Mean Group and partially pooled PVAR estimates.
- ▶ Rebuild a GVAR using alternative trade-weight matrices.
- ▶ Compare GIRFs for shocks originating in different economies.
- ▶ Construct a simple shift-share instrument and discuss whether identification comes from shares or shocks.
- ▶ Add a network visualization based on FEVD or impulse-response spillovers.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 13. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your own countries, regions, policy episodes and networks.
- ▶ Add coding demonstrations in Python, R, MATLAB or Stata.
- ▶ Add alternative weighting matrices and identification schemes.
- ▶ Add applications to trade, climate, sovereign risk or financial spillovers.

Make them yours.

End of the Companion Slides

Chapter 13: Multi-Country Methods for Policy Analysis

From individual economies to an interconnected global system

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